

Theory Building: Rational Choice and Game Theory

POL 51: Scientific Study of Politics

Spring 2026

Three big questions

1. What are models?
2. How can we measure and establish causal relationships (fundamental problem of causal inference)?
3. How likely is an observed outcome a product of random chance?

Is human behavior predictable?

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No, people are weird (have free will).

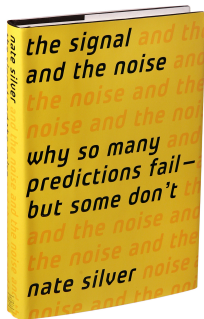


Is human behavior predictable?

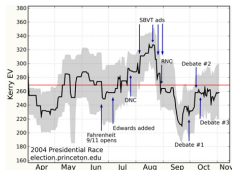
- ▶ What does it matter?
- ▶ Prediction is a big part of science: do we understand the mechanisms that produce observed outcomes?
- ▶ For instance, if we really understand gravity, we should be able to make predictions about it that correspond to what we actually observe (reality).
- ▶ When we evaluate a scientific theory (gravity, evolution, spontaneous generation), we assess its predictive power: what does it say will happen under specific conditions?

Is human behavior predictable?

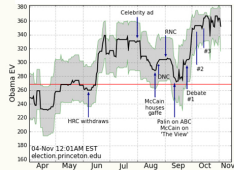
Declining volatility in Presidential campaigns, 2004-2016



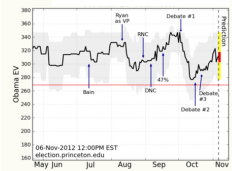
2004: 120 electoral votes



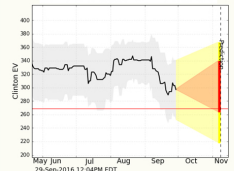
2008: 90 electoral votes



2012: 70 electoral votes



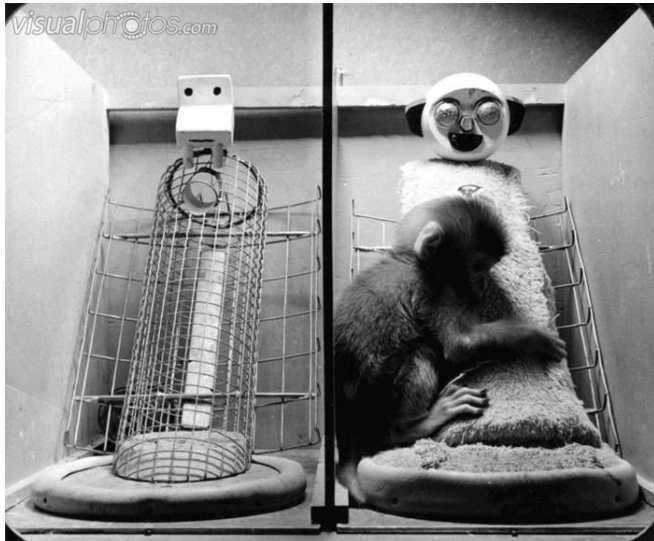
2016: 58 electoral votes so far



The birth of political science

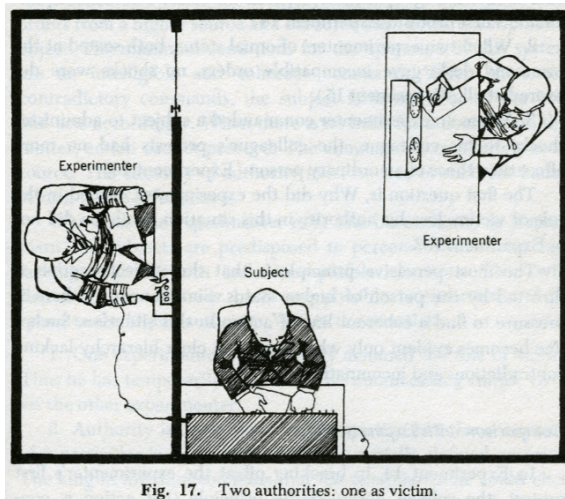
- ▶ Political *science* came of age in the post-war behavioral revolution.
- ▶ **Behavioralism**: the idea that we can study, explain, and predict human behavior just like we do natural phenomena.

Harlow's monkeys



BD5188 [RM] © www.visualphotos.com

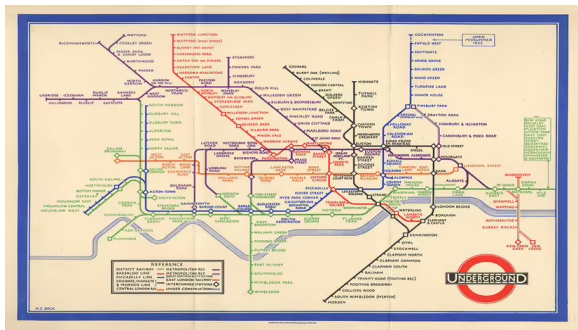
Milgram experiments



Approaching politics scientifically

- ▶ As political scientists, we want to build grand theories (most ambitiously, natural laws) of political behavior.
- ▶ Ultimately, what we want to do is to build *models* of politics.

What makes a good model?

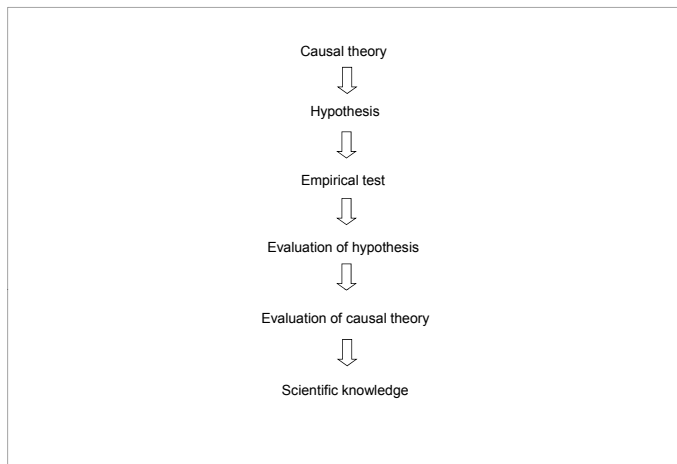


- ▶ Is it useful for what we're trying to do (for example, subway maps)?
- ▶ We want to accomplish generalizability and parsimony (Occam's razor).

Hypotheses

- ▶ To evaluate theories, we develop and test specific hypotheses based on those theories.
- ▶ Hypotheses are statements in the mold of $X \rightarrow Y$.

The road to scientific knowledge



Rational choice theory

- ▶ Grounded in some overarching causal theory that explains observed political events and behavior.
- ▶ For example, rational choice theory:
 - ▶ *Homo economicus*: humans make choices from a set of alternatives that maximize their utility.
 - ▶ How do you decide how to get to class?
 - ▶ Prediction: humans will make the choice that costs them the least and gains them the most. Human behavior is rational.

Game theory

- ▶ Game theory provides a way to model this idea when multiple actors interact (politics) and there is incomplete information (also politics).
- ▶ Game theory makes no sense with chess (there is a solution).
- ▶ Game theory *does* make sense with poker! Bluffing, probabilities, personalities, and uncertainties abound.
 - ▶ Think how you'd model the Cuban missile crisis. Must consider opponent's likely strategy.